

Section 15.7: Triple Integrals in Cylindrical Coordinates

Multiple Integrals

MTH 310
Calculus III

Why Cylindrical Coordinates?

In Section 15.4 we saw that **polar coordinates** simplify double integrals over circular regions. Now we need the same trick for triple integrals over solids with circular symmetry.

Examples of circularly symmetric solids:

- A cylinder $x^2 + y^2 \leq 4$ with $0 \leq z \leq 5$
- A cone $z = \sqrt{x^2 + y^2}$ capped at $z = 3$
- A paraboloid $z = x^2 + y^2$ below $z = 9$

In Cartesian coordinates, the limits involve $\sqrt{\dots}$ expressions from solving $x^2 + y^2 = r^2$. In **cylindrical coordinates**, these become simply $r = \text{const}$ — much cleaner.

Cylindrical coordinates are just **polar coordinates with a z -axis attached**. If the solid has symmetry about the z -axis (or the integrand involves $x^2 + y^2$), cylindrical is the way to go.

The Cylindrical Coordinate System

A point in 3D space can be described by (r, θ, z) where:

- r = distance from the z -axis (same as in polar: $r \geq 0$)
- θ = angle in the xy -plane from the positive x -axis (same as in polar)
- z = height above (or below) the xy -plane (same as in Cartesian)

 A 3D coordinate system showing a point P labeled with cylindrical coordinates (r, θ, z) . The r -coordinate is shown as the horizontal distance from the z -axis to the projection of P in the xy -plane, θ is the angle from the positive x -axis to this projection, and z is the vertical height from the xy -plane to P. Dashed lines connect the components.

Think of it this way: stand at the origin, face along the positive x -axis, turn by angle θ , walk out a distance r — that puts you at $(r, \theta, 0)$ in the xy -plane. Then ride the elevator up (or down) by z .

Cylindrical = polar + height. The "cylindrical" name comes from the fact that $r = c$ is a **cylinder** centered on the z -axis.

Conversion Formulas

Cylindrical $\rightarrow$ Cartesian Conversions

Cylindrical to Cartesian:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z$$

Cartesian to Cylindrical:

$$r^2 = x^2 + y^2, \quad \tan \theta = \frac{y}{x}, \quad z = z$$

Common surface conversions:

Cartesian	Cylindrical	Surface
$x^2 + y^2 = 4$	$r = 2$	Cylinder of radius 2
$z = \sqrt{x^2 + y^2}$	$z = r$	Cone
$z = x^2 + y^2$	$z = r^2$	Paraboloid
$x^2 + y^2 + z^2 = 9$	$r^2 + z^2 = 9$	Sphere of radius 3
$z = 2x$	$z = 2r \cos \theta$	Plane through z -axis

The key substitution is $x^2 + y^2 = r^2$. Any equation involving $x^2 + y^2$ becomes simpler in cylindrical coordinates.

The Volume Element in Cylindrical Coordinates

In Cartesian: $dV = dx dy dz$.

In cylindrical, we need to account for the same Jacobian factor we saw in polar coordinates.

A small cylindrical "wedge" at position (r, θ, z) has:

- Radial thickness: dr
- Arc length: $r d\theta$ (the r factor!)
- Height: dz

So the volume element is:

$$dV = r dz dr d\theta$$

This is exactly the polar area element $dA = r dr d\theta$ with a dz tacked on.

 A small cylindrical volume element showing a wedge-shaped piece with radial thickness dr , arc length $r d\theta$, and height dz . The volume is labeled $dV = r dz dr d\theta$.

Don't forget the extra r ! The factor r appears because the "width" of the wedge in the angular direction is $r d\theta$, not just $d\theta$.

Quick Check: Convert to Cylindrical

Rewrite each equation in cylindrical coordinates:

1. $x^2 + y^2 + z^2 = 9$
2. $x^2 + y^2 = 4$
3. $z = 3x^2 + 3y^2$

Replace $x^2 + y^2$ with r^2 in each equation.

When to Use Cylindrical Coordinates

Use cylindrical coordinates when:

1. **The region** involves cylinders ($x^2 + y^2 = a^2$), cones ($z = c\sqrt{x^2 + y^2}$), or paraboloids ($z = a(x^2 + y^2)$)
2. **The integrand** involves $x^2 + y^2$ (it becomes r^2 — cleaner)
3. **The projection** onto the xy -plane is a circle, annulus, or sector

Template for a cylindrical triple integral:

$$\iiint_E f(x, y, z) dV = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r, \theta)}^{u_2(r, \theta)} f(r \cos \theta, r \sin \theta, z) r dz dr d\theta$$

Typical order: $dz dr d\theta$ (height first, then radial, then angular). But other orders are possible.

If you see $x^2 + y^2$ in the region description or the integrand, think cylindrical.

Example 1: Volume in the First Octant

Example 1

Find the volume of the solid in the first octant that lies under the plane $z = 2x$ and inside the cylinder $x^2 + y^2 = 4$.

Convert the boundaries to cylindrical:

- Cylinder: $x^2 + y^2 = 4 \Rightarrow r = 2$
- Plane: $z = 2x = 2r \cos \theta$
- First octant: $x \geq 0, y \geq 0, z \geq 0$

What are the limits on θ ? On r ? On z ? Don't forget the r in dV !

Example 2: Volume Inside a Cylinder, Below a Cone

Example 2

Find the volume of the solid inside the cylinder $x^2 + y^2 = 4$ between $z = 0$ and the cone $z = \sqrt{x^2 + y^2}$.

Convert the surfaces to cylindrical and set up $\int \int \int r \, dz \, dr \, d\theta$. The full circle means θ goes from 0 to 2π .

Homework

Section 15.7: 3, 9, 13, 17, 19, 23, 25, 31, 33